WBR perception — status report, 14 October 2026

Our next product milestone is a D435i visual-inertial demo on the Raspberry Pi 5.

The artificial intelligence (AI) team reports to you as the project owner.

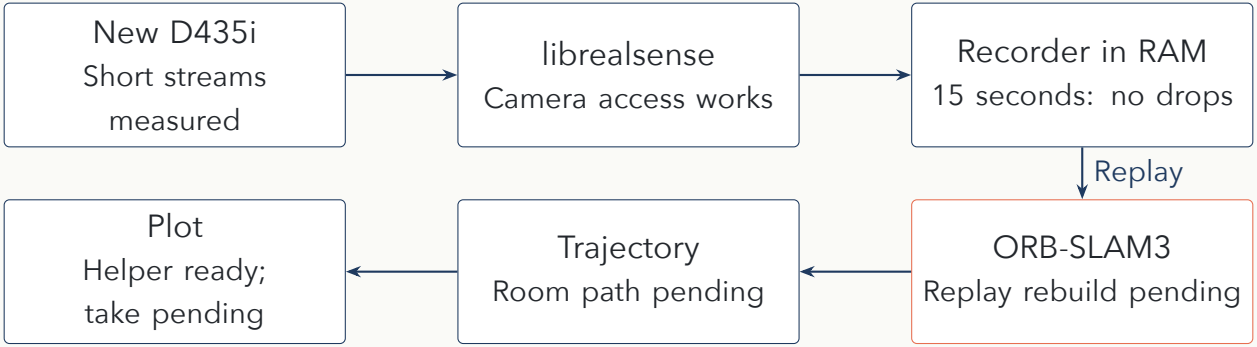
WBR means wheel-biped robot.



D435i

The illustration makes no validation claim.

The input works briefly; the camera-to-plot chain is unfinished.



RAM means random-access memory; SLAM means simultaneous localization and mapping.

The later STM32H7 controller owns balance and motor safety; this demo produces a path only.

Three EuRoC runs match the published accuracy range.

3.7–4.2 cm

Our absolute trajectory error (ATE) was close to the paper's **3.6 cm**.

Relative pose error (RPE) was about **1.3 cm over one second**.

- The public EuRoC machine-hall dataset provides a known reference path.
- The native build reproduces similar accuracy on this sequence.
- This does not prove D435i calibration or real-time speed.
- The 8-gigabyte (GB) Pi rerun is still pending.

Memory recording fixes short-test drops; the long take is pending.

- The 20-second still take delivered both infrared (IR) streams at 30 hertz (Hz) and inertial measurement unit (IMU) data, with a shutdown gap.
- The first 100-second moving take was unusable: 0.3–2.9-second gaps recurred about every five seconds.
- Card writes reached 16.6 megabytes per second (MB/s), below the 18.4 MB/s input; slow storage is the strongly inferred cause.
- Recording into RAM first gave 15 seconds with zero dropped frames and a 33-millisecond maximum gap.

> A 100-second take of about 1.7 GB fits in the 8 GB Pi; it has not yet been measured.

The replay fix needs rebuilding, and lab readiness is unresolved.

- The native build needed three compatibility patches; a fourth fixes the camera replay's first-frame crash from an uninitialized pointer.
- Power loss interrupted that rebuild; the 8 GB Pi boots with the same 32 GB card but still has under-voltage.
- Ethernet is loose and Wi-Fi is unseen on this board; the 128 GB card still needs qualification.
- The 64 GB Universal Serial Bus (USB) stick writes only about 1 MB/s; calibration is also unfinished.

> Next we unblock the lab session, finish the rebuild, and test one complete camera-to-plot route.

First, a prepared lab session makes the remaining tests possible.

- **Why:** power, connectivity and fresh-camera entry checks determine whether we can proceed.
- **You:** confirm the 15 October slot, bring the 8 GB Pi and camera, secure Ethernet or arrange Wi-Fi, and confirm the 5-ampere (A) supply's availability.
- **Team:** check the setup and follow N1, N2 and N3 in order; the mandatory 300-second raw-input soak is retained.
- **Done:** the correct Pi is reachable and entry evidence is reviewed; unmet conditions keep progression blocked.

> At the bench, you must explicitly confirm camera availability before any camera or SLAM execution.

Second, finishing the rebuild lets us retest the crash fix.

- **Why:** camera replay failed before producing a usable trajectory; the applied fix is not yet in a completed build.
- **Team:** resume the interrupted build on the 8 GB Pi and check the example uses the rebuilt library.
- **You:** keep the Pi powered and the lab connection available while the team checks the result.
- **Done:** the build finishes successfully and library linkage is verified, before capture and replay.

Third, a full memory take gives us repeatable camera input.

- **Why:** replaying the same take separates input faults from SLAM faults without repeating every camera movement.
- **You:** move the camera smoothly for 100 seconds, translating within 1–2 seconds, then make a small loop and finish 5–10 seconds still.
- **Team:** record into RAM, copy afterwards, and audit both image streams, motion data, timestamps and power.
- **Done:** recording and copy checks succeed, rates are as expected, and no unexplained gaps remain; otherwise review before replay.

> Keep the Pi and cables fixed; recheck memory and free space before the take.

Fourth, replay and a plot show whether tracking holds together.

- **Why:** offline replay lets us inspect initialization, tracking and resets before a live attempt.
- **Team:** replay the reviewed take on the Pi, then copy and verify the outputs and plot the trajectory on the Mac.
- **You:** review the result with the team before agreeing to the live attempt.
- **Done:** initialization, sustained tracking, one stable map and nonempty trajectories agree with the logs; failures trigger investigation.

> A plausible room plot cannot establish accuracy because there is no reference path.

Fifth, a reviewed live attempt completes the diagnostic demo.

- **Why:** live input tests the final connection from the physical camera to SLAM after replay has been reviewed.
- **You:** remain at the bench and repeat the smooth camera motion, keeping the Pi and connectors stationary.
- **Team:** run the same binary and settings on live input for 90 seconds, inspect tracking and power, then verify shutdown and saved outputs.
- **Done:** retained logs and a trajectory support a one-slide summary of what worked, what failed and what remains unproven.

> Run live only after offline review; record failures as outcomes and keep their evidence.

The demo is a milestone; reliable product behavior comes next.

- **If successful,** the demo shows the D435i-to-trajectory route working on the Pi 5 for the reviewed attempts.
- It does not establish room accuracy, calibrated sensing, sustained real-time speed or motor and jump safety.
- **You provide:** stable 5 A power, the 128 GB card for qualification, and a further lab slot for calibration and repeat tests.
- **Team then verifies:** power and storage, camera–IMU calibration, and fresh N1–N3 evidence before claiming readiness.

> The next decision is lab and hardware availability; acceptance still requires observed results.